

Varenja Jain

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EDUCATION

University of Illinois Urbana-Champaign

2020-2024

Bachelor of Science in Integrative Biology, Minor in Computational Science and Engineering

Urbana, IL

- *Dean's List (Spring 2024)*

EXPERIENCE

IFGC Bioinformatics Scientist, OBGYN Data Analyst

Dec. 2025 - Present

Columbia University Irving Medical Center - Women's Genetics Center

New York, NY

- Analyze fetal ultrasounds and prioritize rare-disease variants into GA4GH Phenopackets to improve differential diagnoses in maternal outcomes and discovery workflows for the Repository of the International Fetal Genomics Consortium
- Implement GA4GH Beacon endpoints at Columbia University and coordinate lightweight Beacon deployments at partner sites such as The Broad Institute and the University of New South Wales
- Construct a database to enable HIPAA-compliant clinical queries of Rare Mendelian diseases across global institutions without bulk data transfer

Research Assistant | Robinson Lab

Jan. 2025 - Nov. 2025

Berlin Institute of Health at Charité — Medical Computer Science and Artificial Intelligence

Berlin, Germany

- Developed a pipeline to transform raw prenatal sonography data into standardized GA4GH Phenopackets
- Spearhead creation of an international federated genotype-phenotype database for the Fetal Sequencing Consortium to study Rare Mendelian diseases
- Wrote software to construct machine-readable, ontology-aware phenotype representations from tabular data sources

Contract Bioinformatician

Dec. 2022 - Nov. 2025

Advanced Genomics Institute and Laboratory Medicine

Paramus, NJ

- Optimized Illumina ICA and BaseSpace workflows, adding SFTP integration to streamline WGS/WES analyses
- Performed targeted enrichment, structural variant detection, and CNV calling to improve clinical interpretation of germline samples
- Utilized Franklin by Genoox database to prioritize causal variants for clinical reporting

Student Researcher

Aug. 2023 - Dec. 2024

Illinois Natural History Survey - Tan Lab of Biodiversity Genomics

Urbana, IL

- **Morphological Diversity Project (2023):**
 - * Researched adaptation in Sisoridae and Amphilidae catfishes within fast-water environments.
 - * Presented findings showing significant shape disparity linked to adhesive organ presence, providing insights into evolutionary constraints on morphology.
- **Phylogenomics Project (2024):**
 - * Investigated ultraconserved elements (UCEs) in zebrafish genomes using low-coverage whole-genome sequencing (lc-WGS) to study evolutionary relationships.
 - * Built and optimized phylogenomic assembly workflows using QC utilities, De Novo Assembly, and the CAPTUS toolkit.

Student Researcher | Metabolomics & Proteomics Core Facilities

June 2023 – Aug. 2023

Roy J. Carver Biotechnology Center

Urbana, IL

- Developed an untargeted metabolomics pipeline for post-processing, Quality Assurance/Quality Control, and data analysis.
- Supported for data processing on LC-MS Untargeted Metabolite Profiling.

SLC Conference Planning Committee | IEEE

Jan 2022 - Jan 2023

The Institute of Electrical and Electronics Engineers

Chicago, IL

- Managed conference logistics, scheduling, and coordinating with stakeholders.
- Worked with Midwest Region 4 Committee members to develop and implement strategies for a successful conference.

SPIN Research Intern | The NEAT Project v4.0

Aug. 2021 – Sep. 2022

National Center for Supercomputing Applications

Urbana, IL

- Developed an NGS toolkit on the HAL cluster, improving parallel processing performance by 7%.
- Implemented alignment algorithms (Smith-Waterman, BLAST, localized string alignment) to enhance mutation and sequencing model pipelines.
- Presented research at the NCSA Exhibition, Engineering Open House, and FoDOMMaT/SPIN Showcase.

Outreach Committee Lead | [Pulse 2022](#)

Aug 2021 - Aug 2022

*Department of Electrical and Computer Engineering**Urbana, IL*

- Coordinated materials and event location logistics with the ECE department during COVID-19.
- Develop Software to exhibit Computer Engineering principles: C++ Data Structures, Polymorphism, Command-Line Interface, Stack/Heap Memory Management, Address Space, etc.
- Design Hardware activities to guide freshmen through simulated Electrical Engineering projects: series vs parallel circuits, Pulse Width Modulation motor control, Thermistor and LDR implementation

Phys 211 Experienced Learning Assistant

Jan 2021 - Dec 2021

*Loomis Laboratory Of Physics**Urbana, IL*

- Continued the study of Physics pedagogy by instructing PHYS 211 labs, providing guidance to 30+ students.
- Answered student questions, clarified critical lecture material, and built/corrected IOLab setups.
- Developed advanced teaching, communication, and leadership skills through interactions with students and seminar presentations alongside lab staff.

IOT Research Lab Assistant | [Caesar Lab](#)

Jan. 2021 – Jun. 2021

*Coordinated Science Laboratory**Urbana, IL*

- Remodeled a Reinforcement Learning System intended for UAV-Assisted Emergency Response.
- Upgraded the fully-distributed communication environment for USAF usage.
- Implemented communication trees via Python Message Passing Interface (MPI) standard in under 6 months.

Research Intern | [Drs. Spitalnik, Hod, La Carpia](#)

June 2018 - Aug 2018

*Columbia University Medical Center - Lab of Transfusion Biology**New York, NY*

- Investigated the effects of transfusional iron overload on gut microbiota due to intravenous infusion.
- Conducted initial studies with a mouse model to retrieve data on iron-deficient erythropoiesis in blood donors and red blood cell recovery after transfusion of hematopoietic red blood transplant.
- Utilized basic Spearman Correlation meta-analysis of bacteria communities, performed blood analysis tests, and used Flow Cytometry to collect sample cell data.

LEADERSHIP AND INVOLVEMENTS

Conferences and Recognitions:

- NCSA Letter of Outstanding Student Leadership Recognition (2023-2024)
- NCSA 2nd Annual Student Research Conference - Planning Committee, Panel Moderator, and Industry Chair (2024)
- ACMG Annual Clinical Genetics Meeting (Metro Toronto Convention Center), 2024
- ISPD 28th International Conference on Prenatal Diagnosis and Therapy (Westin Copley Place, Boston), 2024
- NCSA Student Research Conference - Planning Committee and Panel Moderator (2023)
- IEEE Region 4 - Chicago Nexus Student Leadership Conference - Conference Planning Lead (2022-2023)

Academic Presentations:

- University of Illinois - Undergraduate Research Symposium (2024)
- National Center for Supercomputing Applications - SPIN Lightning Talks (2021-2022)
- National Center for Supercomputing Applications - Engineering Open House (2021)

Teaching:

- ECE PULSE - Outreach Committee Lead (2022)
- Loomis Laboratory Of Physics - "Expert Learning Assistant" for University Mechanics PHYS211 (2021)

Volunteering:

- Shri Sadguru Seva Sangh Trust - Volunteer for breakfast service, educational service, and cow feeding (2024-2025)
- SIGPWNY - Contributed to solving cybersecurity CTF ("Capture the Flag") competitions and developed badge software for FallCTF (2024)
- De Dilse Charitable Inc. - Co-Founder and Lead Volunteer (2019-2022), Member (2023-Present)
- Kaplen Jewish Community Center on the Palisades - Senior Science Counselor and Tikvah Minor Specialty Counselor for Neil Klatskin Summer Camp (2019)
- American Wheat Mission - Volunteer (2018)

Student Organization Memberships: IEEE@UIUC, iRobotics, MRDC, Vex Robotics, ACM

TECHNICAL SKILLS

Computational Genomics: OMIM, UCSC Genome Browser, Ensembl, Illumina Connected Analytics, BaseSpace SequenceHub (DRAGEN), GATK, VCFtools, STAR aligner, BLAST+, phyluce, SPAdes, Captus, Chromosome Analysis Suite (ChAS), genome assembly and annotation

Computational Phenotyping: HPO (Human Phenotype Ontology), Phenopackets, Exomiser, LIRICAL, Phenoboard, PhenoXtract, RareLink, Beacon, fetal biometry (Observer 7, ViewPoint 6)

Biostatistics: RStudio, Principal Component Analysis, Generalized Procrustes Analysis

Wet Lab Experience: Flow Cytometry, PCR, RT-qPCR, Western blotting, recombinant DNA techniques, cell culture (bacterial and mammalian), microscopy (fluorescence, confocal), ELISA, protein purification, immunohistochemistry

Programming Experience: C/C++, CUDA, Python, Rust, Bash

Developer Tools: Git, GitHub, GitLab, Docker, VirtualBox, Kernel-based Virtual Machines (KVMs), Digital Ocean Droplets, VSCode, IntelliJ, Pycharm, RustRover

GNU/Linux: Debian, Fedora, CentOS, AlmaLinux, Raspberry Pi OS, Ubuntu Server, Arch Linux, Manjaro, EndeavorOS

Hardware and Electronics: Breadboards, Raspberry Pi (Zero/3b/4), Arduino UNO, Intel MAX 10 FPGA, oscilloscopes, network analyzers, RLC circuits, operational amplifiers, band-pass filters, Quartus Prime, ModelSim, and SystemVerilog

Research / Professional Practices: Scientific writing and data presentation, familiarity with reproducible research practices, knowledge of FAIR data principles, collaboration in interdisciplinary teams, literature review and hypothesis generation, AGILE software development and project management

PERSONAL PROJECTS

Excel to Phenopacket Converter | [Personal Github](#) July. 2025 – Aug. 2025

- Constructed extensible CLI for working with clinical data and the Human Phenotype Ontology
- Converted existing digital phenotypic records into GA4GH Phenopackets
- Enabled genotype-phenotype database curation by partners at the University of New South Wales

BadgeDev | [SIGPWNY](#) Aug. 2024 – Dec 2024

- Aided in software development for FallCTF participant, sponsor, and staff badges
- Assembled and performed QC on Pico PCB soldering and firmware tests before distribution
- Fixed and replaced badge firmware/hardware in real time during a CTF event

AM Radio | [ECE 210](#) Jan. 2022 – May 2022

- Construct a functioning AM Radio using a Superheterodyne Receiver; Convert digital .wav audio input to analog 3.5mm speaker output
- Utilize Fourier Transforms to convert Time domain signals to Frequency domain responses

Virtual Gloves | [ECE 120 Honors](#) Jan. 2021 – May 2021

- Develop a “virtual keyboard” by moving fingers attached to flex sensors and provide haptic feedback
- Collect data from flex resistors in Arduino Studio and use C++ to send output confirmations to LEDs

LoopKit | [Personal Github](#) July. 2020 – Sep. 2021

- Adapt the LoopKit source code to customize a program for personal monitoring functionalities
- Modify Swift code to circumvent authentication requisites for bolus delivery within the Loop system
- Implement precision adjustments to default carb absorption parameters, optimizing glycemic control dynamics